Milestone 1 Report — Data Quality & Exploratory Data Analysis

Oct 22, 2026 · @shah

Diabetes Risk Miner · DS311 Data Mining · Fall 2026 · Project P05 · Team: Yusra Khosa, Zafeer Rajput, Faizan Bukhari · Dataset: CDC Diabetes Health Indicators (UCI ID 891), BRFSS 2015 survey

Summary

The dataset is technically clean — 253,680 rows, 21 features, no missing values, every value inside its documented range — but four issues need handling before modelling:

1. **Class imbalance:** only 15.8% of respondents have prediabetes or diabetes (5.3 : 1).
2. **Duplicates:** 23,968 rows (9.4%) are exact copies, almost all healthy non-diabetic profiles.
3. **BMI extremes:** BMI goes up to 98; 584 rows above 70 are most likely entry errors.
4. **Skewed day counts:** MentHlth and PhysHlth are 0 for most people, with a spike at 30.

The strongest signals are general health, high blood pressure, BMI, high cholesterol, difficulty walking and age. Lifestyle factors (fruit, vegetables, smoking) matter far less. An untuned logistic regression already reaches ROC-AUC 0.817.

Team contributions and dataset

Each member ran their own standalone Colab notebook; each loads the data from UCI with `fetch_ucirepo(id=891)`.

Member	Role (from proposal)	Notebook	Report section
Yusra Khosa	Data Loading Lead	`notebooks/M1_Yusra_Data_Quality.ipynb`	Part 1 — Data quality
Zafeer Rajput	Modeling Lead (Q1 & Q3)	`notebooks/M1_Zafeer_Imbalance_and_Risk_Factors.ipynb`	Part 2 — Imbalance & risk factors

Member	Role (from proposal)	Notebook	Report section
Faizan Bukhari	Product & Deployment Lead (Q2)	`notebooks/M1_Faizan_EDA_and_Profiles.ipynb`	Part 3 — EDA & patient profiles

Group	Columns	Encoding
Target	Diabetes_binary	0 = no diabetes, 1 = prediabetes or diabetes
Clinical (binary)	HighBP, HighChol, CholCheck, Stroke, HeartDiseaseorAttack, DiffWalk	0 / 1
Lifestyle (binary)	Smoker, PhysActivity, Fruits, Veggies, HvyAlcoholConsump	0 / 1
Healthcare access (binary)	AnyHealthcare, NoDocbcCost	0 / 1
Demographic	Sex (0 = female, 1 = male), Age (13 five-year bands, 18–24 to 80+), Education (1–6), Income (1–8)	integer codes
Continuous	BMI	12–98
Day counts	MentHlth, PhysHlth (bad days in the past 30)	0–30
Ordinal	GenHlth (1 = excellent … 5 = poor)	1–5

Part 1 — Data quality (Yusra Khosa)

1.1 Loading, structure and missing values

Features X are (253,680 × 21) and target y is (253,680 × 1); every column is a whole number. There are **0 missing values in all 22 columns**. The dataset authors had already removed "don't know" and "refused" answers, so the absence of missing values comes from rows deleted upstream — a selection effect noted under limitations.

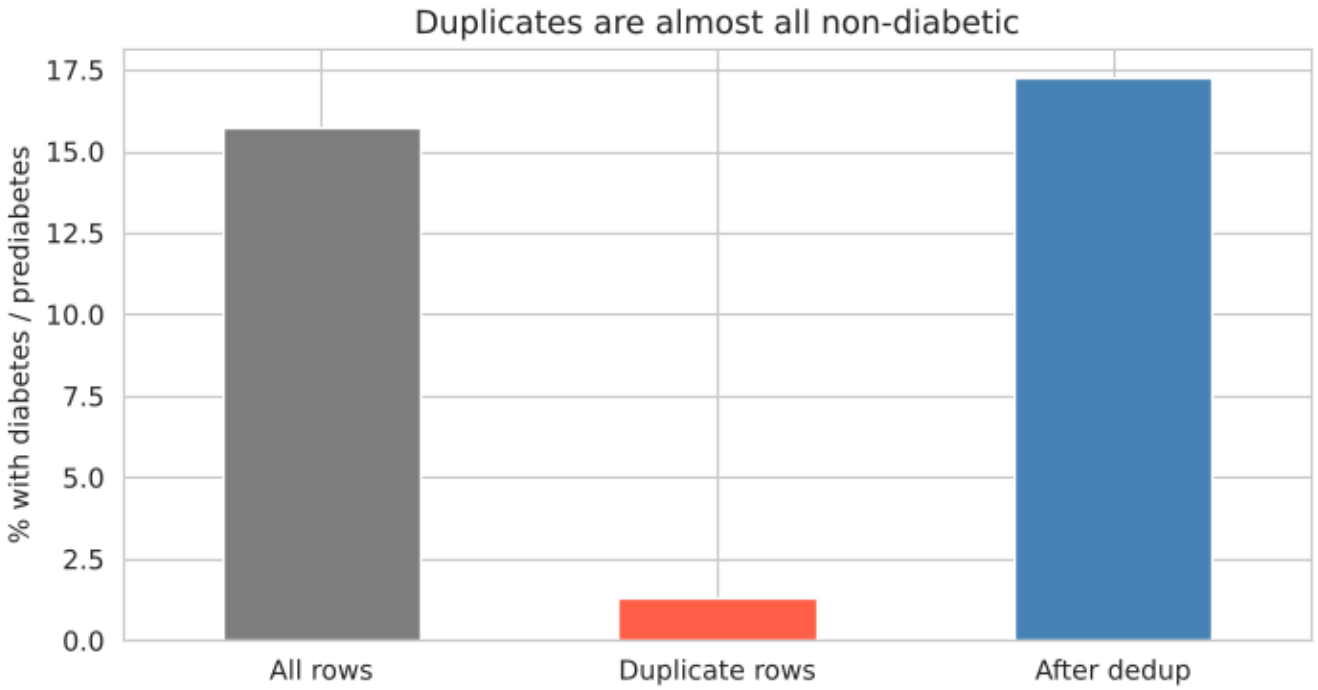
1.2 Validity (range checks)

Check	Rows outside range
GenHlth 1–5, Age 1–13	0, 0
Education 1–6, Income 1–8	0, 0
MentHlth, PhysHlth 0–30	0, 0
14 binary columns only 0/1	0
BMI	12 to 98 (plausible range about 13–70)

Sparse categories: Education = 1 (never attended school) has only 174 rows; Age = 1 (18–24) has 5,700.

1.3 Duplicates and conflicting labels

23,968 rows (9.4%) are exact duplicates, and only 1.3% of them are positive versus 15.8% overall.

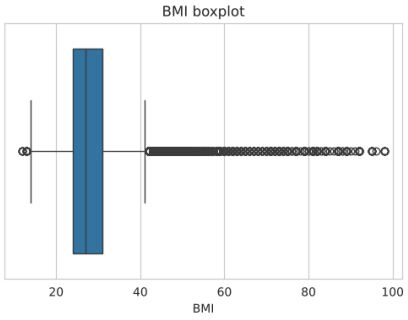


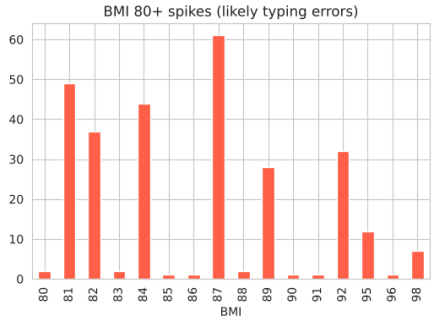
Measure	Value
Exact duplicate rows	23,968 (9.4%)
Typical duplicate	no high BP (81%), no high cholesterol (74%), good health, BMI about 25

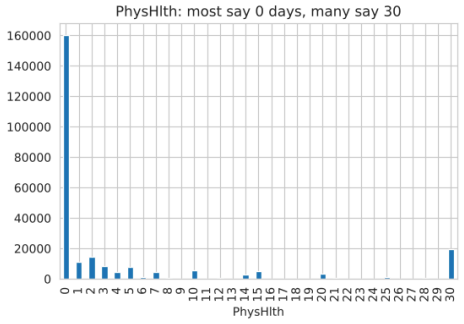
Measure	Value
Rows after removing duplicates	229,712 (positive rate 17.3%)
Identical features with different labels	1,804 feature combinations

With 21 mostly yes/no answers, many healthy people give identical answers, so these are probably different respondents rather than double entries. They must still be removed before splitting: otherwise identical rows land in both train and test and inflate the scores.

1.4 Outliers and skewed columns







Column	Skew	Issue
BMI	2.12	9,847 rows outside the IQR fences (13.5–41.5); 2,547 ⩾ 50; 281 ⩾ 80; 584 > 70; 68 < 15
MentHlth	2.72	69.3% are 0 days, 4.8% are 30 days
PhysHlth	2.21	63.1% are 0 days, 7.6% are 30 days

BMI values of 80 and above pile up on a few numbers (84, 87, 92) instead of tapering off — most likely typing errors. BMI 40–70 is real severe obesity and is kept.

1.5 Consistency checks

Rule	Rows	Decision
High cholesterol but no cholesterol check in 5 years	1,981	Possible (older diagnosis) — keep
"Excellent" health but 30 bad physical-health days	416	Unusual — keep
30 bad physical-health days but no difficulty walking	6,719	Plausible — keep

Rule	Rows	Decision
Age 18–24 with diabetes	99	Plausible (type 1) — keep
No health coverage	12,417	Valid access gap — keep

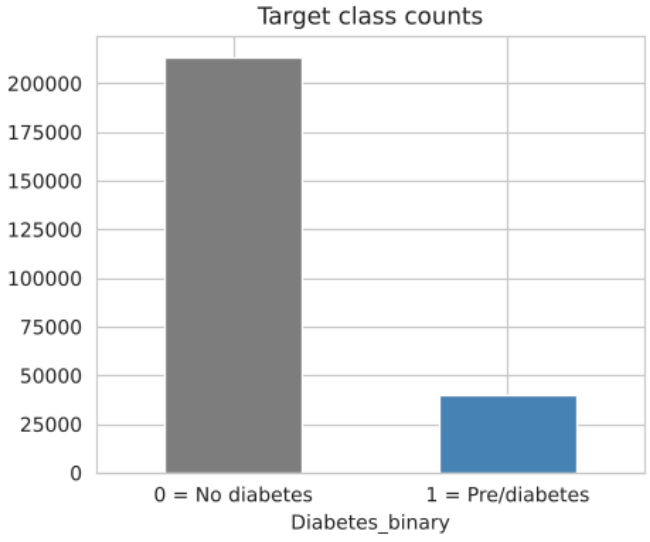
1.6 Cleaning applied

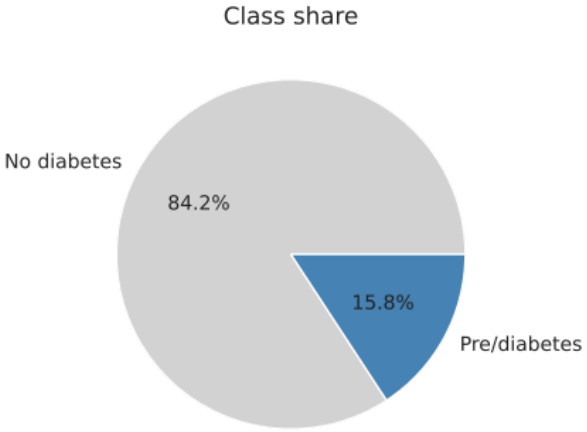
1. Removed 23,968 exact duplicates → 229,712 rows.
2. Capped BMI at 70.
3. Added binned day counts MentHlth_bin / PhysHlth_bin (0 = none, 1 = 1–13 days, 2 = 14–30 days).
4. Merged Education levels 1 and 2.
5. Converted all columns to int16 to save memory.

Part 2 — Class imbalance & risk factors (Zafeer Rajput)

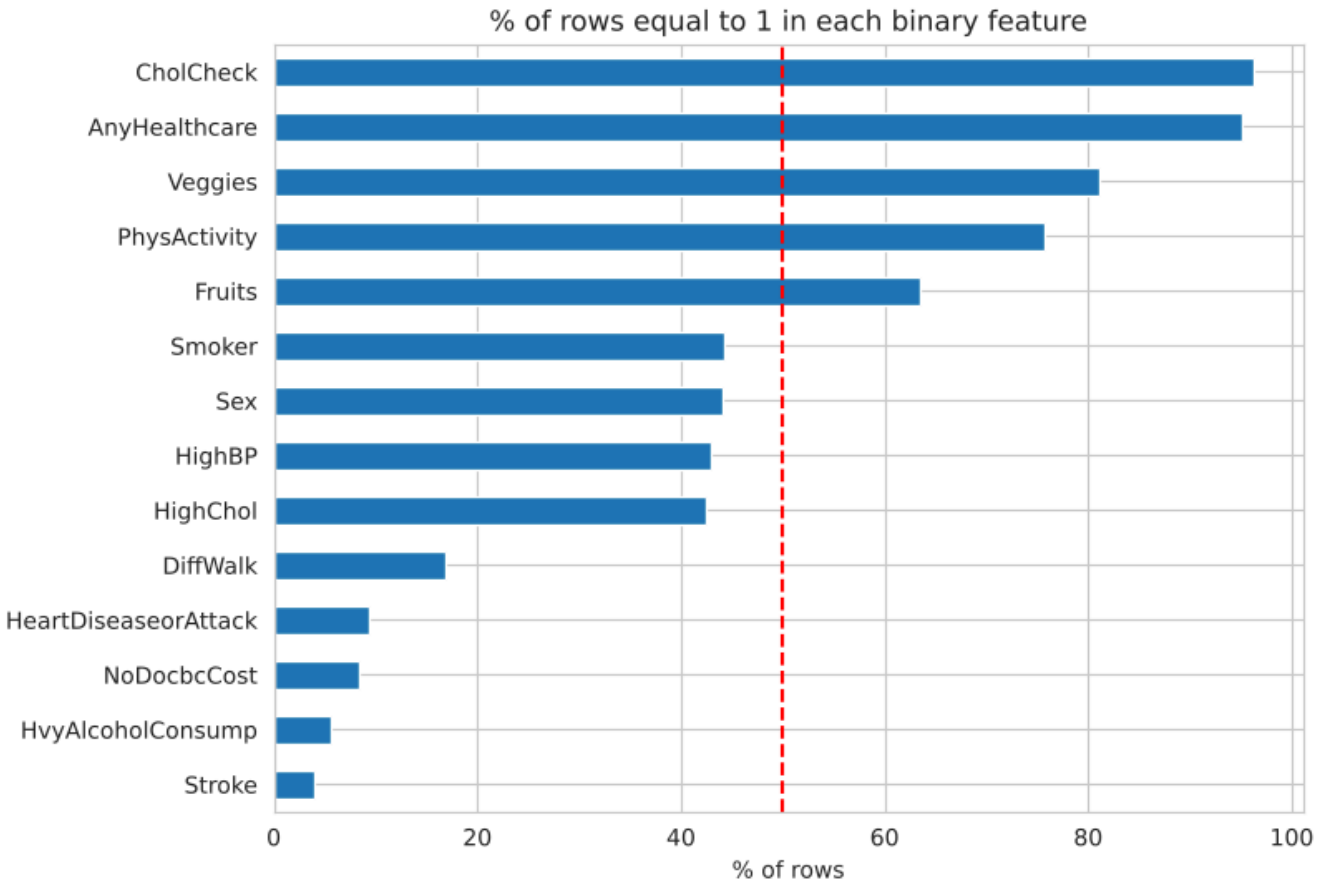
2.1 Class imbalance

Only 15.8% of respondents are positive (39,977 of 253,680), a 5.3 : 1 ratio. A model that always predicts "no diabetes" scores 84.2% accuracy, so the project uses Macro-F1 and ROC-AUC instead of accuracy.



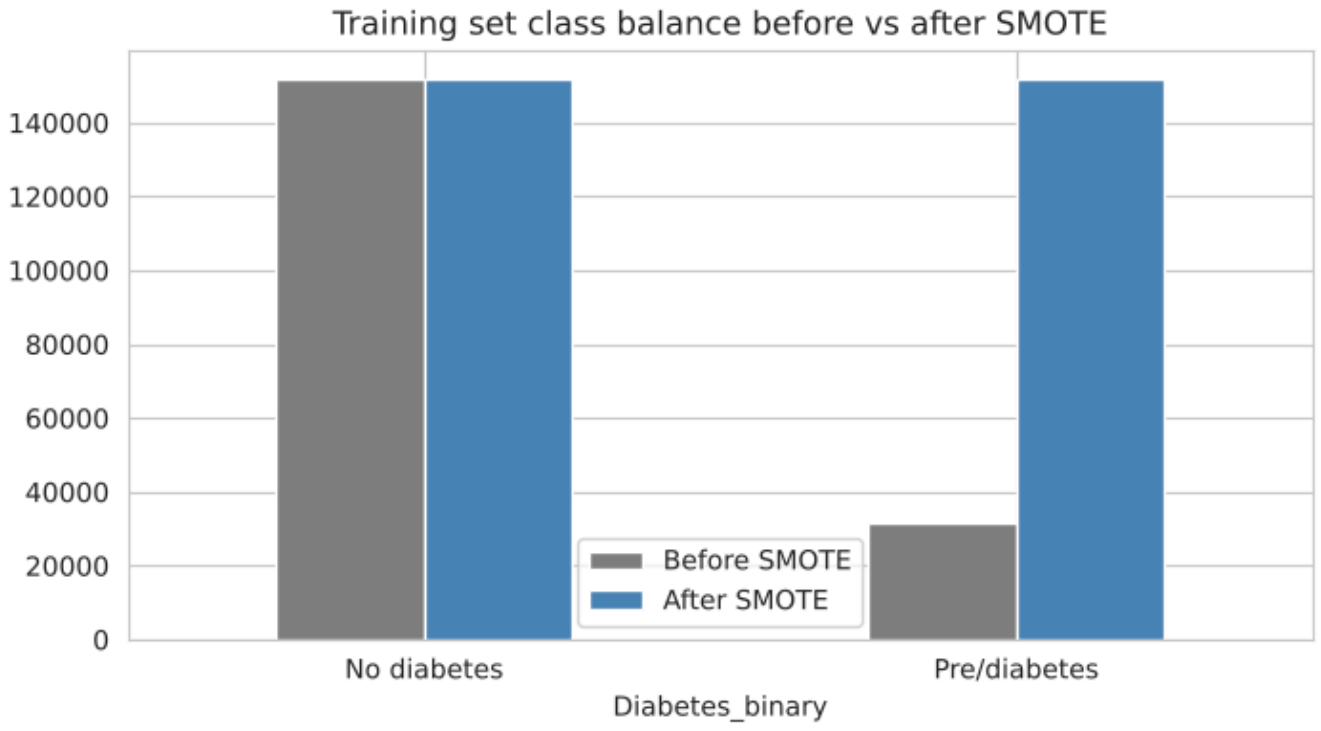


Several features are also very unbalanced: CholCheck (96.3% = 1) and AnyHealthcare (95.1% = 1) carry little information, while Stroke (4.1%) and HvyAlcoholConsump (5.6%) are rare.



2.2 Handling imbalance with SMOTE

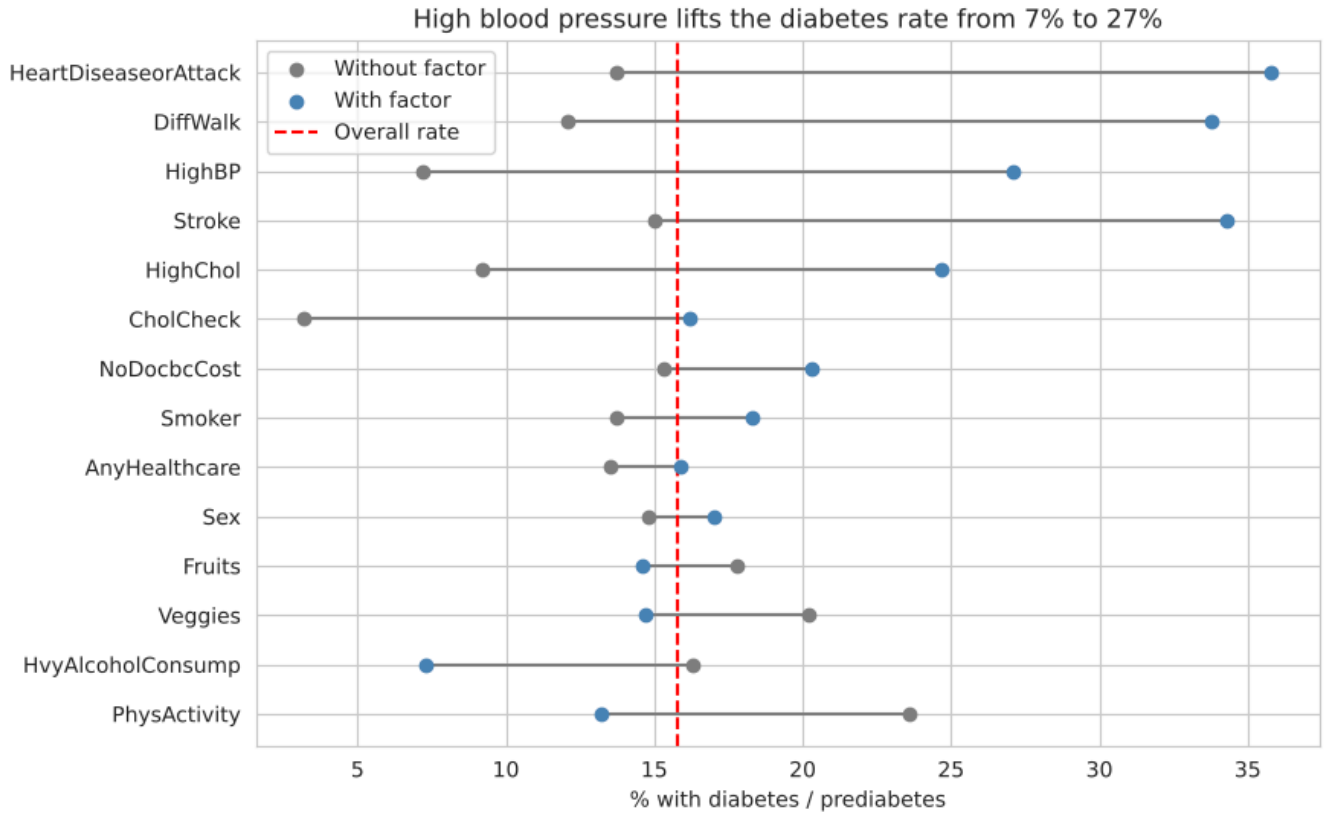
The cleaned data was split 80/20, stratified with random_state = 42, and **SMOTE was applied to the training set only** so that no synthetic rows leak into the test set.



Set	No diabetes	Pre/diabetes
Training, before SMOTE	152,044	31,725
Training, after SMOTE	152,044	152,044
Test (untouched)	38,011	7,932

2.3 Risk factors: diabetes rate with vs without each factor

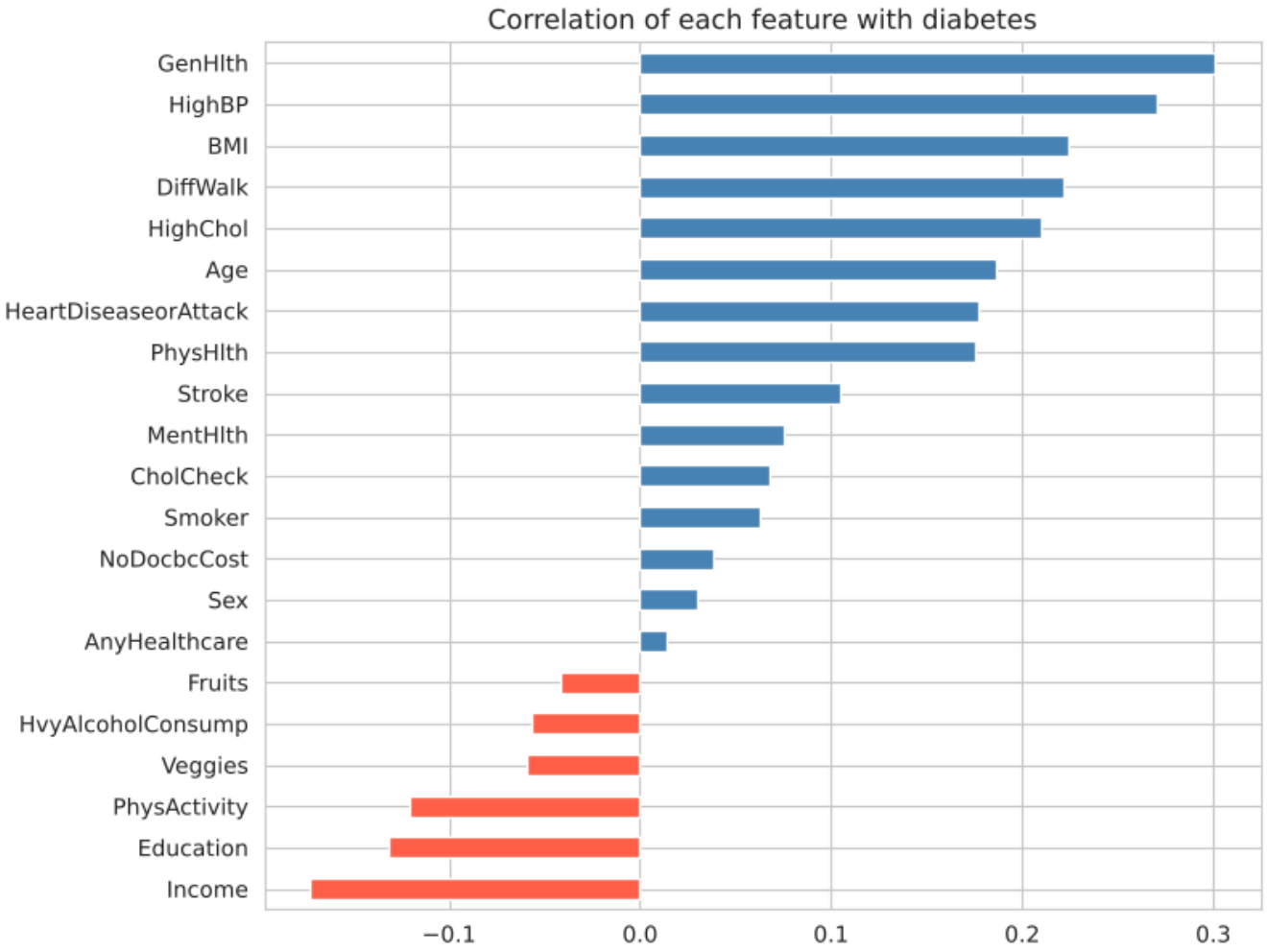
High blood pressure lifts the rate from 7.2% to 27.1%. Clinical conditions each roughly triple the rate; diet and smoking move it by only 3–5 points.



Heavy drinking appearing protective (16.3% → 7.3%) is almost certainly reverse causation: people diagnosed with diabetes are told to stop drinking. The survey is cross-sectional, so all of these are associations, not causes.

2.4 Feature ranking

Correlation and mutual information agree on the top features: GenHlth, HighBP, BMI, DiffWalk, HighChol and Age.

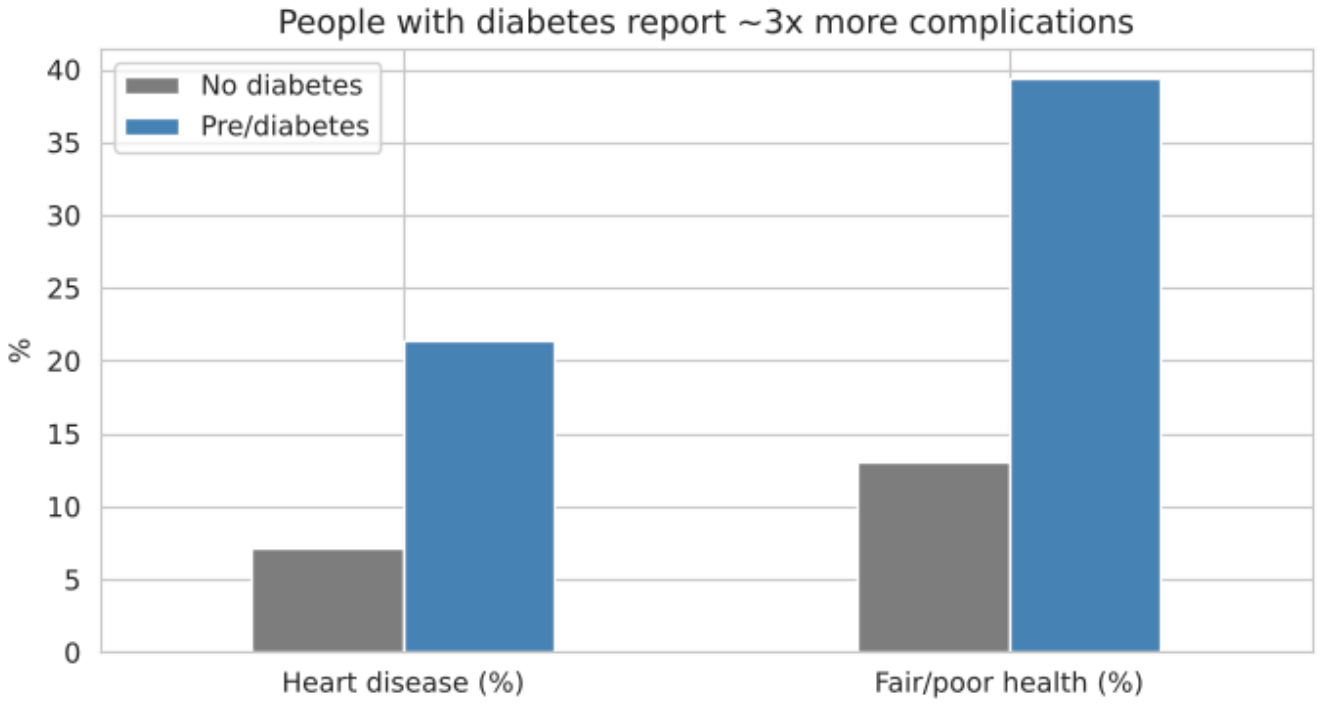


Rank	Feature	Correlation	Mutual information
1	GenHlth	+0.301	0.0462
2	HighBP	+0.270	0.0380
3	BMI	+0.224	0.0310
4	DiffWalk	+0.222	0.0193
5	HighChol	+0.210	0.0240
6	Age	+0.186	0.0226

The weakest features are Fruits, NoDocbcCost, Sex and AnyHealthcare ($|r| < 0.05$).

2.5 Q3 preview: complications by diabetes status

People with diabetes report heart disease three times as often (21.4% vs 7.2%) and fair or poor health three times as often (39.4% vs 13.1%).



When predicting these outcomes in Q3, features that are themselves consequences of the outcome (DiffWalk, PhysHlth, Stroke) should be left out to avoid leakage.

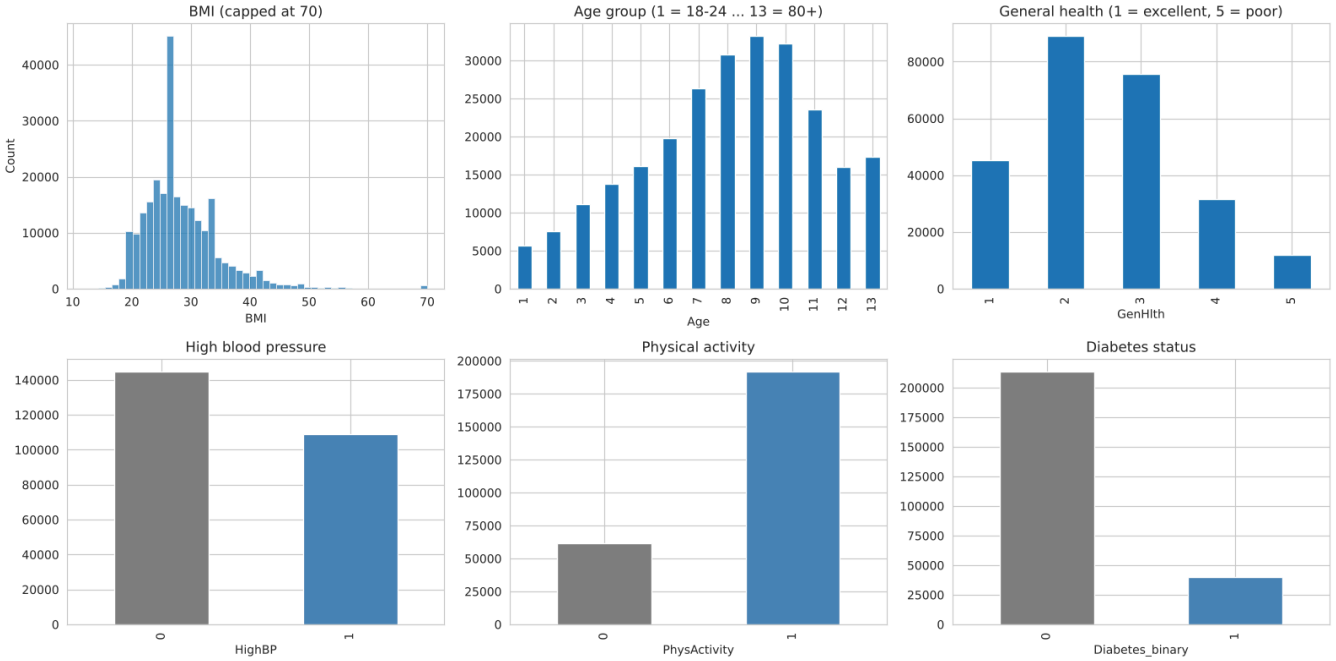
2.6 Baseline (5-fold stratified cross-validation)

Model	Score
Majority-class baseline	Macro-F1 0.457
Logistic regression (untuned, all features)	ROC-AUC 0.817

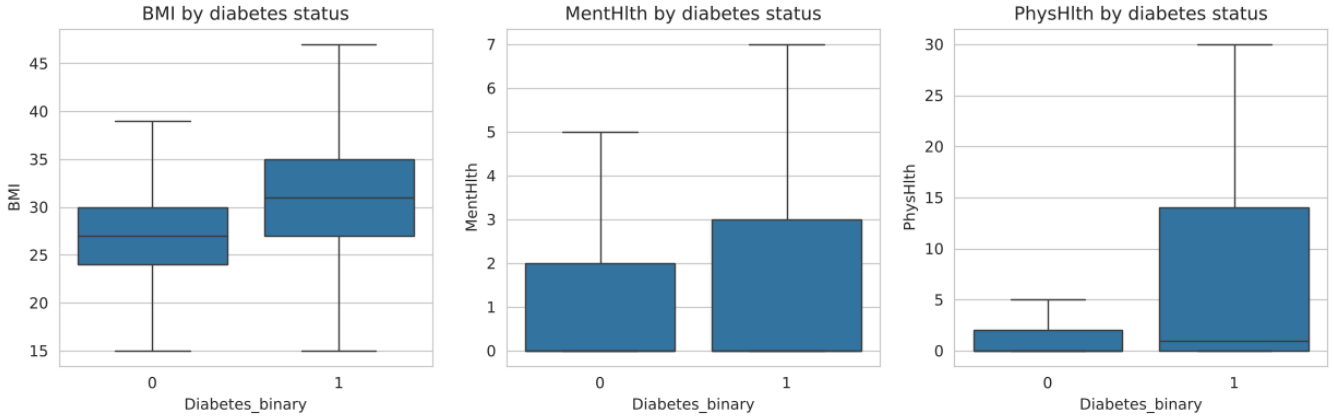
Part 3 — EDA & patient profiles (Faizan Bukhari)

3.1 Distributions of key indicators

Most respondents are 45–69 years old and rate their health "very good" or "good"; about 43% have high blood pressure and 76% are physically active.

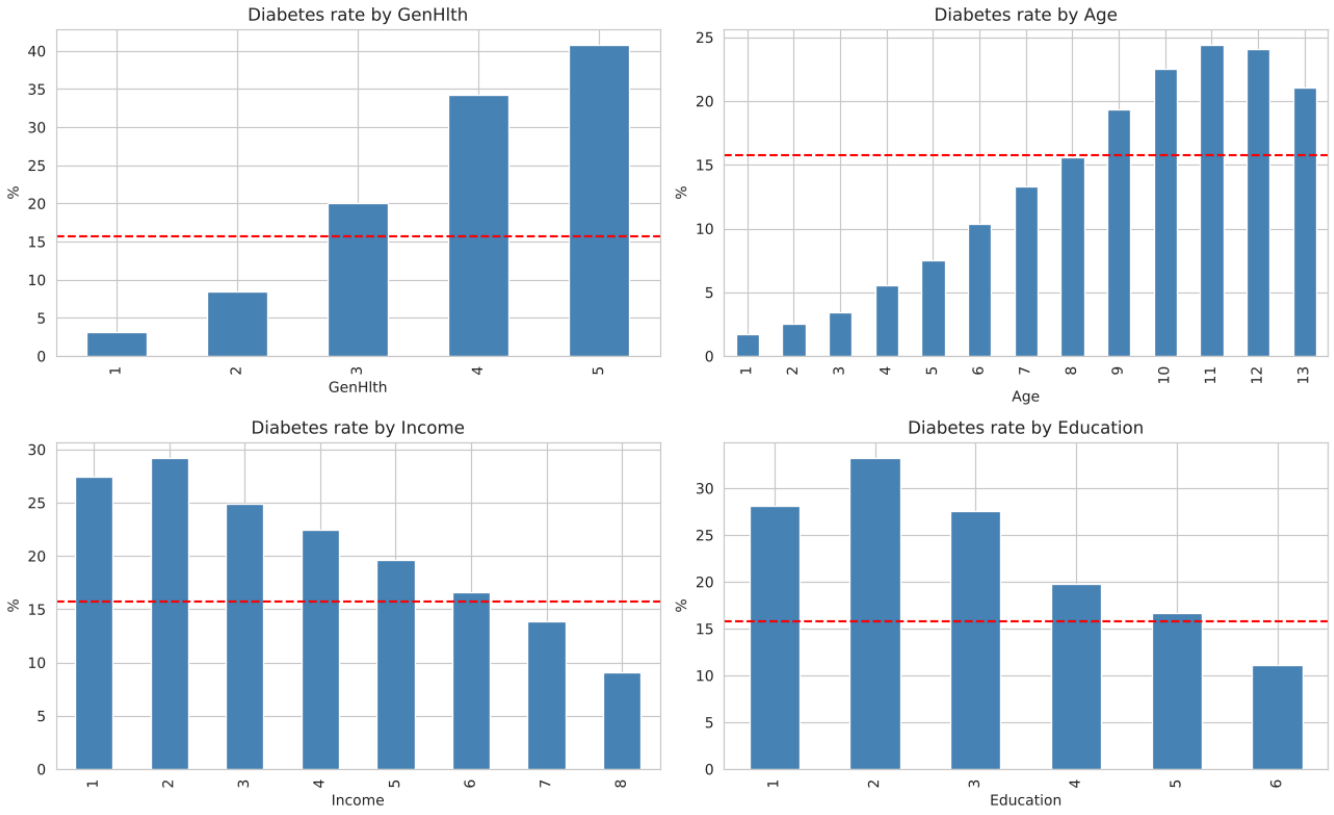


People with diabetes have a higher BMI and report about twice as many bad physical-health days.



Group	Mean BMI	Mean MentHlth (days)	Mean PhysHlth (days)
No diabetes	27.7	2.9	3.6
Pre/diabetes	31.8	4.5	7.8

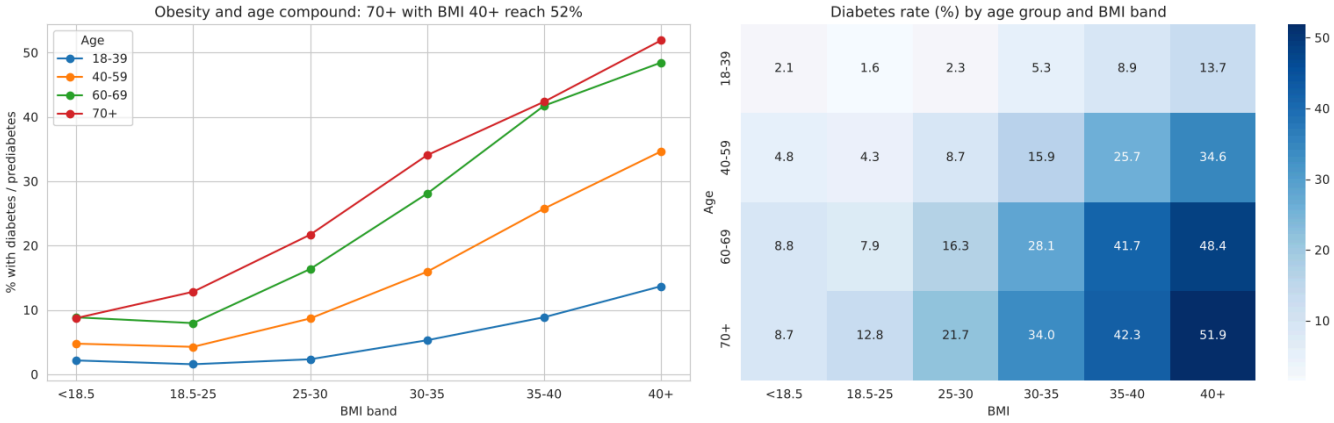
3.2 Diabetes rate across ordinal features



- **General health:** from 3.2% (excellent) to 40.8% (poor) — the strongest single signal.
- **Age:** rises steadily to about 24% at ages 70–79, then dips slightly at 80+.
- **Income:** falls from 27.5% in the lowest band to 9.1% in the highest.
- **Education:** falls from about 28–33% (none or elementary) to 11.1% (college graduates).

3.3 BMI and age together

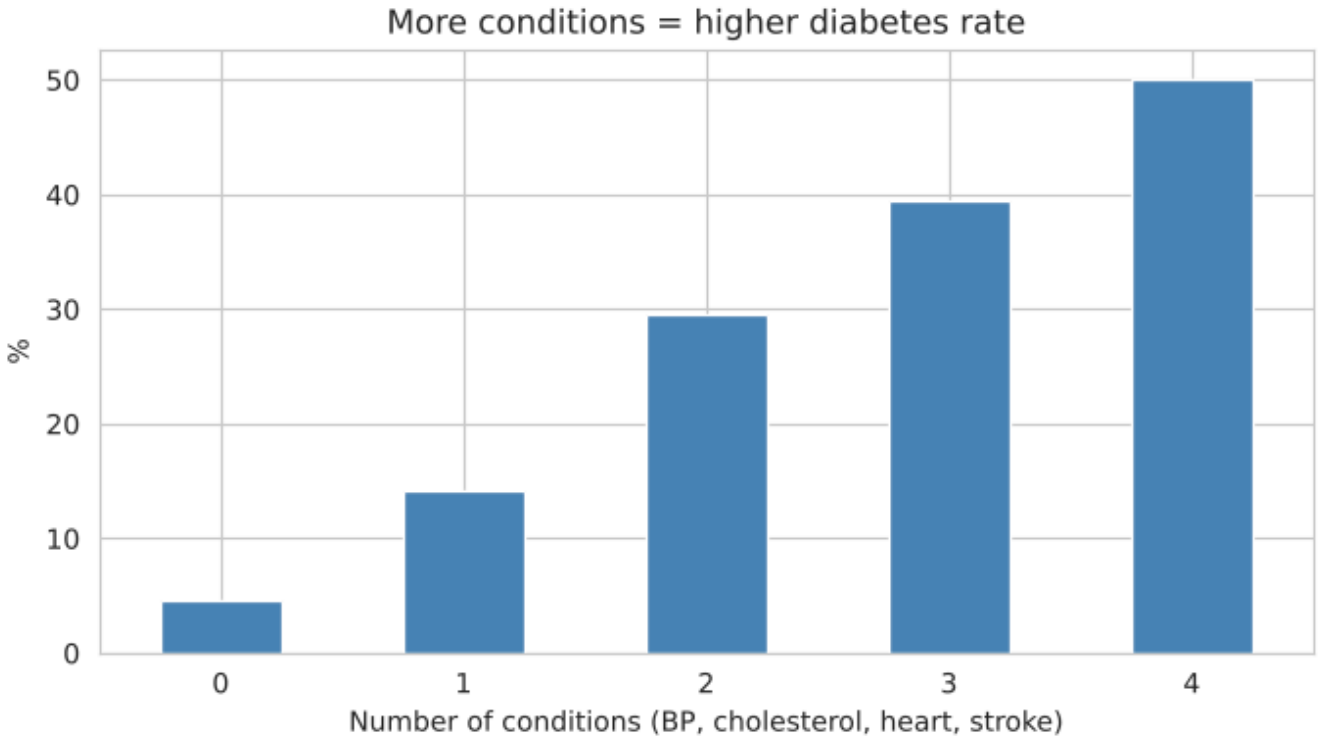
BMI and age compound: more than half (51.9%) of people aged 70+ with BMI 40+ have diabetes or prediabetes, against 1.6% of under-40s with normal BMI.



The lines fan out as BMI rises, so the effect of BMI is stronger in older people — an interaction that tree models capture automatically and logistic regression needs as an extra term.

3.4 Comorbidity profiles

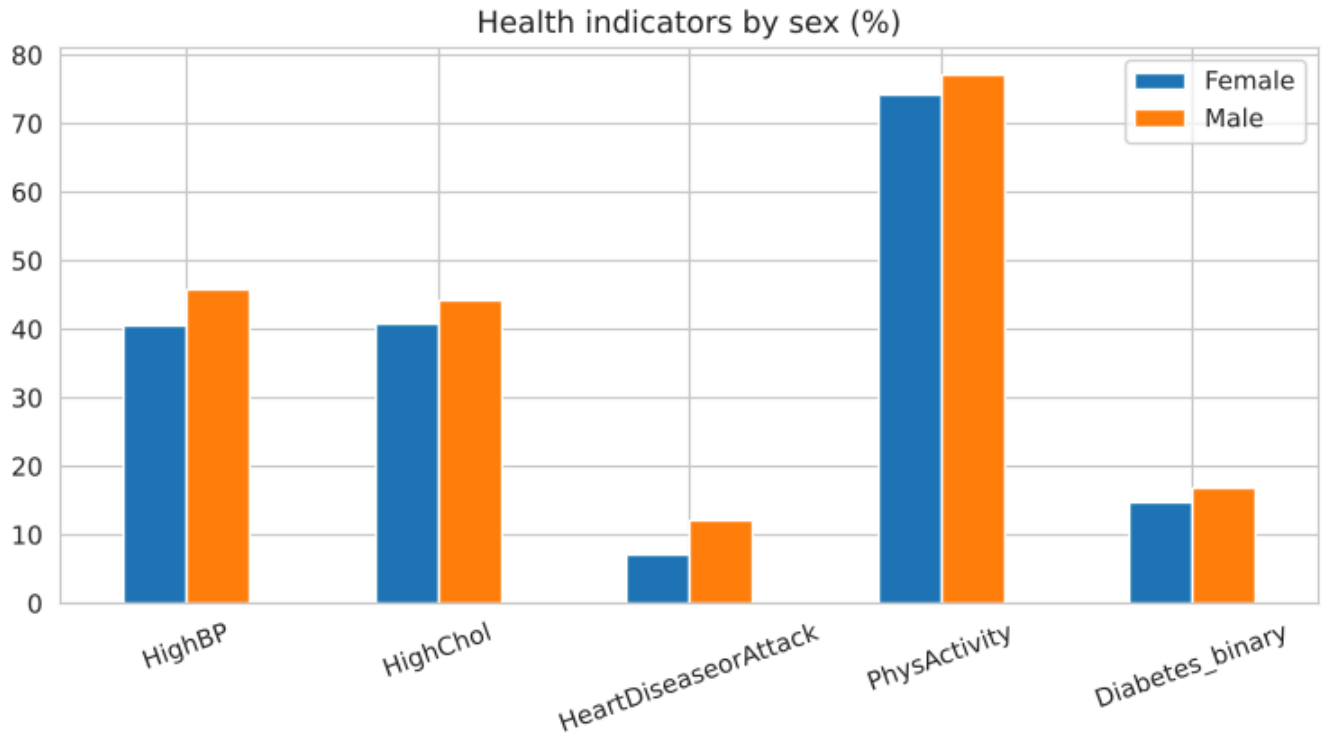
The diabetes rate climbs from 4.5% with none of the four conditions (high BP, high cholesterol, heart disease, stroke) to 50.1% with all four.



Each additional condition adds roughly 10–15 points, making this count a strong engineered feature and a natural axis for the Q2 clusters.

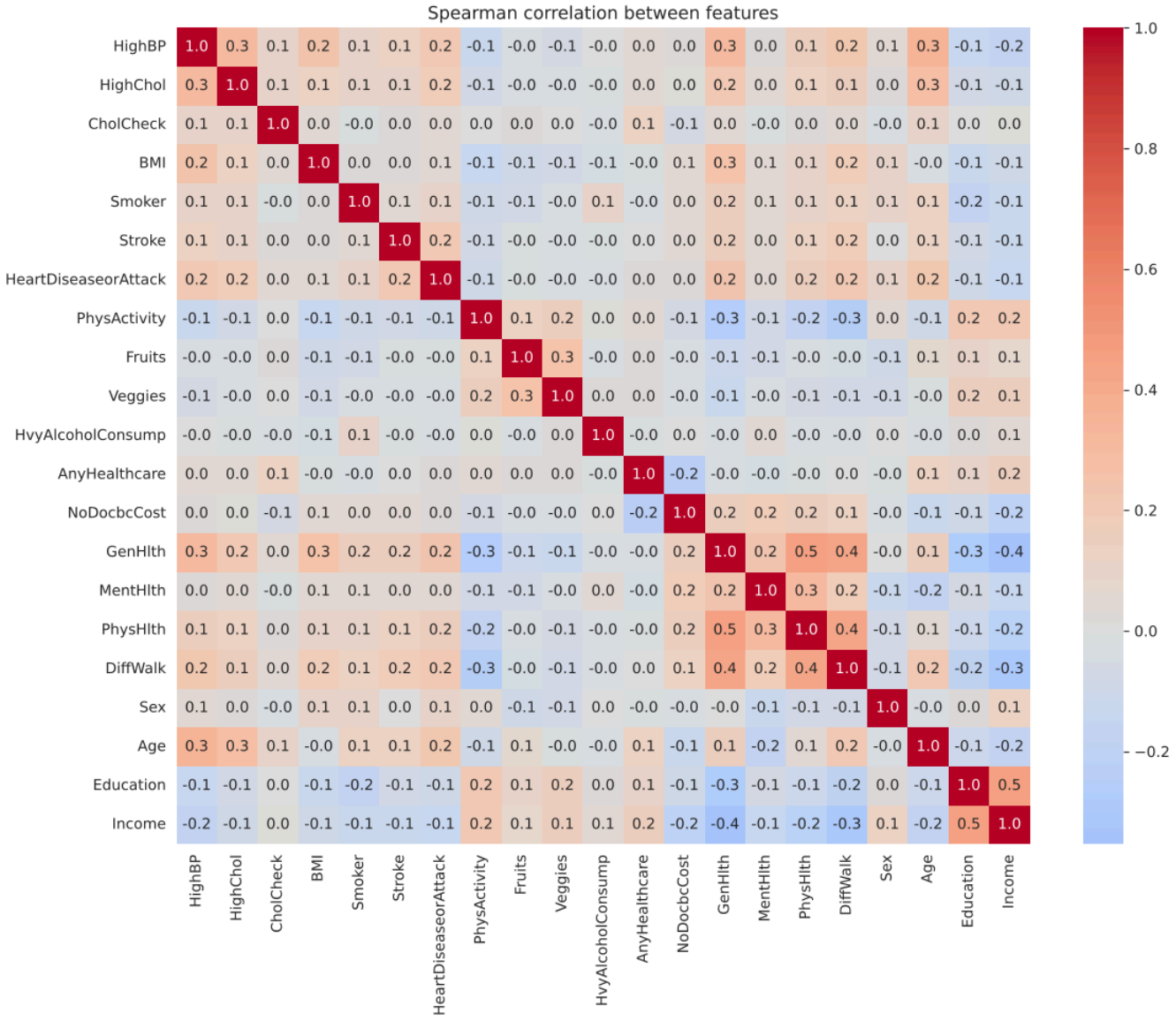
3.5 Demographic breakdown by sex

Men are slightly more often positive (17.0% vs 14.8%) and report heart disease more often (12.3% vs 7.2%).



3.6 Feature relationships (input for clustering, Q2)

No two features are strongly redundant: the highest Spearman correlation is 0.45.



Related features fall into three groups that clusters are likely to follow:

- **Physical health burden:** GenHlth–PhysHlth 0.45, GenHlth–DiffWalk 0.42, PhysHlth–DiffWalk 0.42
- **Socioeconomic status:** Education–Income 0.45, GenHlth–Income −0.35
- **Cardiometabolic / ageing:** HighBP–Age 0.35, HighBP–GenHlth 0.30, HighBP–HighChol 0.30

14 of the 21 features are binary, and BMI (std 6.6) and the day counts (std 7–9) are on much larger scales than the 1–5 codes. Features must therefore be scaled before PCA/K-Means, and duplicates removed so they don't form an artificial "healthy" cluster.

What this means for Milestone 2

1. Use the cleaned data: duplicates removed, BMI capped at 70.

2. Split stratified on the target with a fixed seed; apply SMOTE or class weights to the training fold only.
3. Evaluate with Macro-F1 and ROC-AUC (PR-AUC for Q3) against the majority-class baseline.
4. Expect ROC-AUC around 0.82, since the strongest features are only moderately correlated with the target ($|r| \leqslant 0.30$).
5. Scale features before PCA/K-Means for Q2.

Limitations and how to reproduce

- Self-reported survey data (BRFSS 2015), so BMI and diagnoses may be inaccurate.
- Cross-sectional data: findings are associations, not causes.
- Prediabetes and diabetes are merged into one positive class.
- Rows with "don't know" or "refused" answers were removed by the dataset authors before release.
- Results are for education and exploration only, not medical diagnosis.

To reproduce, open any of the three notebooks in the repository's `notebooks/` folder (github.com/yusrakhosa/diabetes-risk-miner) in Google Colab and choose Runtime → Run all. Each installs ucimlrepo, downloads the data from UCI and runs independently.